

MASTER TRADING INTELLIGENCE HANDBOOK

Knowledge foundation for a private AI crypto trading agent

GOVERNING PRINCIPLE

NO TRADE is a first-class decision and is often preferable. The agent must never manufacture trades. No setup is claimed profitable. Facts, heuristics, hypotheses, and system rules are kept distinct.

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Retrieve by concept name. Part numbers exist only for human navigation.

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0. How to Use This Handbook

This document is an internal knowledge layer for a private AI crypto trading agent. It is not a retail course and it is not a strategy catalog. Its purpose is to define what market objects mean, how they interact, when they are usable as evidence, when they fail, and what would strengthen or weaken a thesis.

The agent must retrieve individual sections. Every major concept is written to stand alone. When a section is retrieved, treat missing context as a reason to remain conservative, not a reason to invent a narrative.

0.1 Classification system

Every claim in this handbook belongs to one of four classes. The agent must not collapse them.

- **FACT** — Market mechanics, definitions, identities, or documented venue rules. These can be treated as operational premises after verifying they still match current venue documentation.
- **HEURISTIC** — A commonly used interpretation or rule of thumb. It may be useful as a prior. It is not proof. It must be tested against the agent's own data.
- **HYPOTHESIS** — A possible explanation or relationship that is contested, incomplete, or context-dependent. Never treat as established.
- **SYSTEM RULE** — A constraint on the agent's own decision process. These are process requirements, not market truths.

SYSTEM RULE

The agent must never manufacture a trade to satisfy activity, completeness, or narrative pressure. NO TRADE is a first-class output and is often the correct output. A low-quality trade is worse than inaction.

0.2 Retrieval keys

When querying this handbook, retrieve by concept name, not by chapter number. Useful keys include: definition, calculation, information content, common interpretation, failure mode, interaction, required data, NO-TRADE condition, FACT/HEURISTIC/HYPOTHESIS/SYSTEM RULE.

0.3 What this handbook does not contain

- No claimed win rates, Sharpe ratios, or historical performance of any setup.
- No invented academic citations, papers, or statistics.
- No promise that any setup is profitable.
- No instruction to force a directional view when evidence is incomplete.
- No substitute for live venue documentation, contract specs, or fee schedules.

AUDIT NOTE

Venue rules, fee tiers, leverage caps, funding formulas, oracle composition, and API fields change. Part 24 labels Hyperliquid items that must be re-verified against current official documentation before use in production.

1. Market Fundamentals

Financial markets are mechanisms that match buyers and sellers and produce a transaction price. In continuous electronic markets, that matching is usually an order book: a ranked list of resting limit orders. Price is not a metaphysical fair value. Price is the last agreed exchange, plus the current best bid and best offer that define where the next trade can occur.

1.1 How financial markets work

Definition. A market is a set of rules plus a matching engine (or dealer) that converts orders into trades. Participants submit orders that specify instrument, side, size, and constraints (limit price, time-in-force, reduce-only, post-only, etc.). The engine matches compatible orders and publishes resulting trades and book updates.

Price discovery is the process by which new information, inventory needs, hedging demand, and liquidity provision are compressed into a quoted bid/ask and a last trade. Discovery is not the same as correctness. A market can discover a price that later looks wrong. That is still discovery.

FACT

In a central limit order book, a trade occurs only when an aggressive order crosses the spread and matches resting liquidity. Mid-price is not a trade. Mark price is not a trade. A candle close is an aggregation of trades (or of last prices), not an independent object.

Spot vs perpetual futures

Spot is the purchase or sale of the asset itself (or a claim that is intended to deliver or represent the asset). After a spot buy, the account holds the asset and no derivative expiry or funding obligation is inherent to the spot position itself.

Perpetual futures (perps) are derivatives with no expiry. They track an underlying index through a funding mechanism and through arbitrage. A long perp is a leveraged (or unleveraged) claim on the change in contract price, collateralized by margin. It is not ownership of coins.

Because perps can be long or short and can use leverage, their microstructure includes funding, mark price, liquidation, open interest, and basis. Spot does not have those objects in the same form.

Long vs short

A long profits if price rises after entry (ignoring costs). A short profits if price falls. In spot markets, a short usually requires borrowing the asset. In perps, a short is native: the contract is two-sided. Shorting a perp does not require locating borrow, but it does require margin and it does pay or receive funding.

FAILURE / MISUSE

A short is not 'the opposite of a long' in cost space. Funding, borrow, liquidation distance, and squeeze dynamics are asymmetric. Do not assume that a short setup is valid merely because the inverse long setup would be valid.

Market orders vs limit orders

A market order requests immediate execution against available resting liquidity. It is aggressive. It pays the spread and any walk-the-book slippage. A limit order specifies the worst acceptable price. If it rests, it is passive and may earn the spread. If it is marketable (already crossing), it behaves like a constrained market order.

Time-in-force variants matter: GTC rests until filled or canceled; IOC fills immediately and cancels the remainder; FOK requires full immediate fill or cancel; post-only rejects if it would take. Reduce-only prevents increasing position size.

Bid, ask, spread

The bid is the highest price a resting buyer will pay. The ask (offer) is the lowest price a resting seller will accept. The spread is ask minus bid. $\text{Mid} = (\text{bid} + \text{ask}) / 2$. Spread is a direct transaction-cost component for takers and a compensation component for makers.

$\text{spread_abs} = \text{ask} - \text{bid}$ | $\text{spread_bps} = 10000 * (\text{ask} - \text{bid}) / \text{mid}$

Slippage

Slippage is the difference between the expected execution price (often mid, last, or limit) and the price actually received. Sources include spread, finite depth, competing orders, latency, and adverse selection. Quoted slippage from a depth snapshot is an estimate, not a guarantee, because the book can change before the order arrives.

Liquidity, depth, order books

Liquidity is the ability to transact size at low cost with limited price impact. It is not synonymous with volume. A market can print high volume while being fragile, and a market can be deep at the touch while being thin a few ticks away.

Market depth is the quantity available at each price level. An order book is the data structure of those levels. L1 is best bid/ask. L2 is aggregated size per price. L3 is individual order-level data (not always available). Retail systems often have L1 plus OHLCV and sometimes L2 snapshots.

HEURISTIC

Tighter spreads and thicker near-touch depth usually imply lower expected impact for small orders. This is a cost heuristic, not a directional signal.

Price discovery

Discovery occurs when informed or urgent flow hits the book and remaining quotes adjust. In crypto, discovery is fragmented across many venues. A move on one venue can be imported to another through arb and through shared mark/oracle construction. Fragmentation means 'the' price is a family of related prices: last, mid, mark, index, oracle.

Volatility and volume

Volatility is the dispersion of returns over a window. It is not direction. High volatility widens the distribution of both wins and stop-outs. Volume is transacted size over a window. Volume confirms that trades occurred; it does not by itself say who was informed.

$\text{realized_vol_window} \approx \text{stdev}(\log \text{ returns}) * \text{sqrt}(\text{periods_per_year})$ [annualized convention varies]

Realized volatility is computed from observed returns. Implied volatility, where an options market exists, is the volatility parameter that makes an option model match observed option prices. Most perp-only retail stacks do not observe implied vol directly. Do not invent it from candles.

Leverage, margin, liquidation

Leverage is notional exposure divided by equity allocated to the position (isolated) or to the account (cross). Initial margin is the equity required to open. Maintenance margin is the equity required to keep the position open. Liquidation is the venue's forced reduction or closure when equity falls below maintenance, using mark price (on venues that define mark).

$\text{notional} = \text{position_size} * \text{contract_price}$

$\text{effective_leverage} \approx \text{notional} / \text{equity_attributed}$

FACT

Liquidation is an execution event under venue rules, not a technical-analysis pattern. The liquidation price depends on entry, size, margin mode, maintenance parameters, funding, and other positions in cross margin. Displayed liq prices can change after entry.

Funding, open interest, mark / index / oracle

Funding is a periodic payment between longs and shorts designed to tether perp price to an index/oracle. Positive funding typically means longs pay shorts. Negative means shorts pay longs. Exact sign convention and formula are venue-specific.

Open interest (OI) is the outstanding size of open contracts. It is not volume. Volume can rise while OI falls if positions are being closed. OI can rise with little price change if both sides are adding.

Index price is usually a basket of external spot prices. Oracle price is the robust external reference a venue uses for funding and sometimes as a mark input. Mark price is the venue's fair-perp estimate used for margining, uPnL, liquidations, and often TP/SL triggers. Mark can diverge from last trade during thin or violent books.

How these variables interact

Price, depth, and volatility determine impact. Impact plus spread plus fees plus funding determine round-trip cost. Leverage maps price moves into margin consumption. Crowded leveraged positioning plus thin books can produce liquidation cascades: forced selling or buying that consumes resting liquidity and creates further mark moves. Funding is both a carry cost and a crowding symptom. OI tells you whether exposure is expanding or contracting, not who will be right.

WHEN USEFUL	WHEN UNRELIABLE	WHAT NOT TO INFER
Use these objects to compute cost, capacity, liquidation distance, and whether a proposed size is executable.	Do not treat funding, OI, or spread as standalone directional oracles. They are state variables.	Do not infer 'smart money' from a single print, a single funding print, or a single OI change.

Object	What it is	Primary use	Common misuse
Last / mid	Trade or quote snapshot	Mark-to-market of fills; quote state	Treating mid as guaranteed fill
Spread / depth	Cost and capacity	Size and order-type choice	Reading depth as future direction
Volume	Transacted size	Activity / impact context	Equating volume with conviction
Realized vol / ATR	Observed dispersion	Stop distance, regime	Equating vol with trend
Leverage / margin	Amplification + constraint	Risk budget, liq distance	Using max leverage as default
Funding	Carry + tether mechanism	Hold-cost, crowding context	Always fade or always follow funding
OI	Open contract stock	Position expansion/contraction	Assuming OI rise is bullish
Mark / oracle	Venue reference prices	Liq, uPnL, funding base	Assuming mark = executable last

SYSTEM RULE

Before any trade thesis, compute: executable size at acceptable impact, fees + expected slippage + expected funding over intended hold, distance to liquidation at planned leverage, and whether invalidation is inside the noise of current ATR. If those cannot be computed, output NO TRADE.

2. Price Action and Market Structure

Price action is the raw sequence of trades aggregated into bars. Market structure is a language traders use to describe that sequence. Some structure terms are objectively definable given a rule. Many are discretionary labels. The agent must keep that distinction explicit.

2.1 Objectively definable vs subjective

Concept	Can it be made objective?	Requirement
OHLC candle	Yes	Specified interval and aggregation rule
Swing high/low	Yes, if parameterized	Need N-bar pivot rule or zigzag threshold
HH / HL / LH / LL	Yes, given swings	Depends on the swing definition
Trend / range label	Partly	Needs a classification rule; boundaries are arbitrary
Support / resistance	Partly	Needs a detection rule; 'strength' is interpretive
Supply / demand zone	Mostly subjective	Zone geometry is discretionary unless coded
Liquidity pool	Heuristic / inferred	Stops are not fully visible on OHLCV
BOS / CHOCH	Yes, if parameterized	Must freeze the swing algorithm first
Liquidity sweep	Inferred	Requires a definition of prior high/low and reclaim
Failed auction	Interpretive	Auction theory mapping is a model, not a fact

SYSTEM RULE

Do not treat a discretionary label as evidence. If a structure concept is used in a setup, it must have a coded definition, including lookback, pivot length, and tie-break rules. If the definition is not coded, the concept cannot be a trigger.

2.2 Candlesticks

A candle is a compression of trades over an interval into open, high, low, close (and usually volume). Wick geometry is a function of the extrema inside the interval, not a drawing of intent. The same sequence of trades can produce different candle shapes under different interval alignments.

FAILURE / MISUSE

Do not assign narrative meaning to single candle names (doji, hammer, engulfing) without a tested rule. Pattern names are labels. They are not mechanisms.

2.3 Swings and trend grammar

A common objective swing rule: a swing high is a bar high with N strictly lower highs on both sides; a swing low is a bar low with N strictly higher lows on both sides. N is a parameter. Changing N changes the entire structure map.

Given swings: higher high (HH) is a swing high above the prior swing high; higher low (HL) is a swing low above the prior swing low; lower high (LH) and lower low (LL) are the bearish pair. An uptrend grammar is HH+HL. A downtrend grammar is LH+LL. A range often mixes them.

HEURISTIC

Trend-following setups usually want structure agreement with the intended side. Mean-reversion setups often want a mature excursion inside a defined range. Neither grammar predicts the next tick.

2.4 Ranges, consolidation, compression, expansion

A range is a period where price oscillates between identifiable bounds without persistent HH+HL or LH+LL. Consolidation is a range after a directional move. Compression is declining range width / realized vol (ATR falling, bands tightening). Expansion is a rise in range width and often volume after compression.

Compression is a volatility state. It does not encode direction. Expansion is also a volatility state. Breakout systems try to trade expansion; mean-reversion systems try to fade failed expansion.

2.5 Breakouts, false breakouts, retests

A breakout is a close or trade beyond a predefined boundary. A false breakout is a breakout that returns inside the boundary within a defined window and invalidates the break condition. A retest is a return to the broken boundary from the far side.

HYPOTHESIS – NOT ESTABLISHED FACT

A successful retest is sometimes treated as acceptance of new value. This is auction-theory language. It is a model of behavior, not a law. Many retests fail.

2.6 Support, resistance, supply/demand

Support / resistance are prices where opposing interest previously appeared with enough force to pause or reverse a move. They can be coded as prior swing levels, session highs/lows, opening range, VWAP bands, or volume-profile nodes. Supply / demand zones extend those levels into rectangles based on impulse origin. Zone width is usually discretionary.

FAILURE / MISUSE

A level that 'worked' historically is not obligated to work again. Levels are crowded maps. Crowding can make a level hold (resting interest) or fail violently (stop run).

2.7 Momentum, exhaustion, continuation, reversal

Momentum is the persistence of returns over a short window (rate of change, consecutive closes in one direction, displacement vs ATR). Exhaustion is a heuristic that a move has traveled far relative to its recent distribution (extension from VWAP/mean, RSI extremes, waning volume on new extrema). Exhaustion is not a clock.

Continuation means the next impulse is expected in the prior direction after a pause. Reversal means the next impulse is expected opposite. These are setup families, not states that can be known in advance.

2.8 Market structure shift, BOS, CHOCH

In coded structure systems, a break of structure (BOS) is a break of the most recent swing in the trend direction (e.g., breaking prior HH in an uptrend). A change of character (CHOCH) is a break against the prior trend grammar (e.g., breaking a HL in an uptrend). Naming varies by author.

These labels inherit every weakness of the swing algorithm: they flip in chop, they lag, and they can be fitted after the fact by choosing N.

2.9 Liquidity language

Traders call clusters of stops above equal highs / below equal lows liquidity pools. Equal highs/lows are objectively detectable if a tolerance (ticks or ATR fraction) is set. Previous day/week highs and lows and session highs/lows are objective once the session calendar is defined. Crypto sessions are conventions (Asia/London/New York hours), not exchange closures.

A liquidity sweep is a heuristic: price trades through a prior high/low (where stops may rest) and then reclaims back through that level. A failed auction is related language: price probes a new area and is rejected, leaving a short-lived excursion.

FACT

OHLCV cannot see resting stop orders. Sweep language is an inference from price path, not an observation of the stop book. L2 can show visible resting size but still cannot show hidden stops or liquidation thresholds of other accounts.

WHEN USEFUL	WHEN UNRELIABLE	WHAT NOT TO INFER
Use structure as a map of where the market has already turned or paused, and as a source of coded invalidation levels.	Choppy regimes produce many BOS/CHOCH flips. Low-liquidity wicks create fake swings.	Do not infer that liquidity 'must' be taken. Do not infer intent of 'the market maker' from a wick.

3. Technical Indicators

Indicators are deterministic transformations of price and/or volume. They do not contain information that was not already in the input series, except insofar as the transform highlights a feature (average, dispersion, imbalance of up/down volume). They are not independent sensors.

SYSTEM RULE

The LLM must not compute indicators, invent levels, or 'eyeball' a crossover from text. Code computes indicators. The LLM may only interpret structured values that code already produced.

SYSTEM RULE

Do not double-count correlated signals. RSI, stochastic, MACD histogram, and short-rate-of-change often describe the same momentum feature. SMA(50), EMA(50), and VWAP-distance can describe the same location-vs-mean feature. Combine families, not clones.

3.1 Shared failure modes

- Lag: any average of past prices is late at turning points.
- Whiplay in ranges: crossovers and oscillator mid-line crosses fire often without follow-through.
- Parameter sensitivity: 14 vs 21 vs 9 can invert the story on the same chart.
- Regime mismatch: trend tools bleed in ranges; mean-reversion tools die in trends.
- Lookahead in construction: using future bars, wrong shift, or close-of-bar signals on incomplete bars.

SMA — Simple Moving Average

1. What it measures. The arithmetic mean of the last N closing prices (or another specified series).
2. How it is calculated. $SMA_t(N) = (C_t + C_{t-1} + \dots + C_{t-N+1}) / N$. Equal weight on each of the N observations.
3. Information content. A smoothed location of recent price. Contains no volume, no path except the sampled series, no future.
4. Common interpretations. Price above SMA = bullish bias; below = bearish. Rising SMA = uptrend. Crosses of two SMAs as signals.
5. When interpretations fail. Fails in ranges (repeated crosses). Lags in fast moves. A close across an SMA is not a regime change.
6. Combination notes. Use as a location feature (distance from SMA / ATR), not as a second vote alongside EMA of similar length.

EMA — Exponential Moving Average

1. What it measures. A weighted mean of the entire history with exponentially decaying weights, dominated by recent prices.
2. How it is calculated. $EMA_t = \alpha C_t + (1-\alpha) EMA_{t-1}$, with $\alpha = 2/(N+1)$ in the common convention. Initialization method matters.
3. Information content. Similar to SMA but more reactive. Still only a transform of the chosen price series.
4. Common interpretations. Same family of trend and crossover interpretations as SMA, with faster response.
5. When interpretations fail. Faster response means more false flips. Seed/warmup effects can distort early values after a gap or listing.
6. Combination notes. Treat EMA(N) and SMA(N) as substitutes, not confirmations. Pair with a volatility and a structure feature.

VWAP — Volume-Weighted Average Price

1. What it measures. The average price of transactions weighted by volume, usually reset on a session or day boundary.

2. How it is calculated. $VWAP = \sum(P_i * V_i) / \sum V_i$ over the reset window. Intraday implementations often use typical price * bar volume as an approximation when tick data is absent.
3. Information content. A volume-weighted mean for the session. Institutional execution desks use it as a benchmark. It is not a support law.
4. Common interpretations. Price above VWAP = buyers in control of the session; fade back to VWAP; use bands of $k * \text{session stdev}$.
5. When interpretations fail. Reset choice dominates. 24/7 crypto has no official session; a UTC midnight VWAP is a convention. Thin volume makes VWAP jumpy. It is not predictive by construction.
6. Combination notes. Good as an intraday location and execution benchmark. Do not pair with another session-mean of the same window.

RSI — Relative Strength Index (Wilder)

1. What it measures. A bounded oscillator of recent average gains versus average losses.
2. How it is calculated. $RS = SMMA(\text{gains}, N) / SMMA(\text{losses}, N)$; $RSI = 100 - 100/(1+RS)$. Default N is 14. Wilder smoothing is not the same as EMA unless implemented carefully.
3. Information content. A rescaling of recent directional return mass. It does not measure 'strength of the asset' in any fundamental sense.
4. Common interpretations. $RSI > 70$ overbought, < 30 oversold; divergence between price extrema and RSI extrema as reversal warning.
5. When interpretations fail. In strong trends RSI can stay 'overbought' or 'oversold' for long periods. Divergence is easy to overfit and often fails in trend continuation. Bounded oscillators look extreme precisely when trends are healthy.
6. Combination notes. Use as one momentum coordinate. Do not also count stochastic and CCI as separate confirmations of the same overbought label.

MACD

1. What it measures. The gap between a fast and a slow EMA of price, plus a smoothed version of that gap.
2. How it is calculated. $MACD \text{ line} = EMA(\text{fast}) - EMA(\text{slow})$ (common 12, 26). $Signal = EMA(MACD, 9)$. $Histogram = MACD - Signal$.
3. Information content. A band-pass view of price: it emphasizes swings between two smoothing lengths. It is price, twice-smoothed.
4. Common interpretations. Signal-line cross; zero-line cross; histogram waning as loss of momentum.
5. When interpretations fail. Classic lag plus range whipsaw. Histogram 'loss of momentum' occurs in almost every pullback, including those that continue.
6. Combination notes. MACD is a cousin of dual-MA trend systems. Do not add an EMA cross on the same lengths as a second system vote.

ATR — Average True Range

1. What it measures. Typical range of price movement, including gaps, over N bars.
2. How it is calculated. $True \ Range = \max(H-L, |H-C_{prev}|, |L-C_{prev}|)$. ATR is Wilder SMMA of TR, commonly N=14.
3. Information content. A volatility / noise scale. Directionless by design.
4. Common interpretations. Stop distance = $k * ATR$. Regime filter: ATR high vs its own median. Position size inverse to ATR.
5. When interpretations fail. ATR rises after a shock, so stops placed as $k*ATR$ immediately after a spike can be wide just when risk should be reduced — or appropriately wide because noise increased. The interpretation is a policy choice, not a fact.
6. Combination notes. Primary risk-scaling feature. Complementary to trend/momentum features rather than a duplicate.

Bollinger Bands

1. What it measures. A moving average envelope at $\pm k$ standard deviations of price.
2. How it is calculated. $\text{Mid} = \text{SMA}(N)$, commonly 20. $\text{Upper/Lower} = \text{Mid} \pm k * \text{stdev}(N)$, commonly $k=2$. The stdev window and SMA window should be specified.
3. Information content. Location of price versus a recent mean, scaled by recent dispersion. Contains the same information as z-score of price.
4. Common interpretations. Tagging the upper band is overbought; squeeze (band width low) precedes a big move; walk-the-band is a trend.
5. When interpretations fail. The 'squeeze precedes a big move' claim is a volatility-regime statement and does not give direction. In trends, price can ride the band. Mean-reversion fades of band tags fail in expansion regimes.
6. Combination notes. Band width is a volatility feature; band %B is a location feature. Do not also treat z-score of close vs SMA as a separate confirmation.

Stochastic oscillator

1. What it measures. Where the close sits inside the N-bar high-low range.
2. How it is calculated. $\%K = 100 * (C - L_N) / (H_N - L_N)$. %D is a moving average of %K. Lookback and smoothing are parameters.
3. Information content. A range-position oscillator. In a persistent trend the close stays near the range extreme, so the oscillator saturates.
4. Common interpretations. Crosses of %K/%D; overbought/oversold thresholds (80/20); divergence.
5. When interpretations fail. Same trend-saturation problem as RSI. Very noisy on short N. Highly correlated with RSI on many series.
6. Combination notes. Choose RSI or stochastic as the bounded-momentum feature, not both, unless the system has tested incremental value.

ADX – Average Directional Index

1. What it measures. The strength of directional movement, not its sign.
2. How it is calculated. Built from +DM and -DM (up-move vs down-move), normalized by TR, smoothed, then $\text{DX} = 100 * |+\text{DI} - -\text{DI}| / (+\text{DI} + -\text{DI})$. ADX is a smoothing of DX. Common $N=14$.
3. Information content. A trend-strength coordinate. +DI vs -DI carries sign; ADX itself does not.
4. Common interpretations. ADX rising / above 20–25 means a trend exists; fading ADX means the trend is dying.
5. When interpretations fail. Thresholds are arbitrary. ADX can rise in a violent range. ADX turns with lag. A falling ADX does not imply a safe fade.
6. Combination notes. Useful as a regime feature alongside ATR and structure. Do not treat ADX high as permission to ignore invalidation.

Volume, OBV, and related

1. What it measures. Volume is transacted size. On-Balance Volume adds volume on up closes and subtracts it on down closes. Other variants: VWMA, accumulation/distribution, volume delta if tick classification exists.
2. How it is calculated. $\text{OBV}_t = \text{OBV}_{t-1} + V_t * \text{sign}(C_t - C_{t-1})$ with a convention for unchanged closes (often add 0).
3. Information content. Volume measures activity. Signed-volume accumulators measure a crude imbalance of activity on up vs down bars. They do not identify aggressive buyers unless you have trade-side classification.
4. Common interpretations. Rising price + rising volume confirms; divergence of OBV vs price warns of reversal.

5. When interpretations fail. Crypto volume is fragmented across venues and can be inflated or inconsistent. Up/down bar classification is a blunt proxy for aggressor side. Divergence is easy to cherry-pick.
6. Combination notes. Prefer venue-specific volume for execution context. Treat cross-venue 'total volume' as approximate. Do not stack OBV + A/D + VWAP slope as three independent volume truths.

3.2 Moving-average crossovers

A crossover is a state change of two smoothers. It is a lagging classification of recent drift. It will always look brilliant on a past trend and expensive in a past range. It is not a third source of information beyond the two averages.

3.3 Volatility measures

Use a small family: ATR or realized stdev for scale; band width or realized-vol percentile for compression/expansion; range/ATR for bar expansion. Implied vol belongs only if options data is actually present.

WHEN USEFUL	WHEN UNRELIABLE	WHAT NOT TO INFER
Indicators are useful as features: location, momentum, volatility scale, trend strength, activity.	They are unreliable as standalone entry oracles, especially when several of the same family agree.	Do not infer an edge from an indicator reading. An edge is a tested mapping from features to outcomes after costs.

4. Order Flow and Derivatives

Order flow describes how orders arrive and interact with the book. Derivatives state describes how leveraged positioning is distributed and paid for. Together they can add context to price. They do not replace price, and they are not deterministic triggers.

4.1 Book objects

Bid/ask depth is resting size at or near the touch and beyond. Order-book imbalance is typically $(\text{bid_size} - \text{ask_size}) / (\text{bid_size} + \text{ask_size})$ over the first K levels. Spread is the immediate round-trip cost at size ≈ 1 tick. Market depth is the function mapping size to expected average fill.

HEURISTIC

Persistent imbalance at the touch sometimes precedes a short-horizon tick in that direction. The effect, when it exists, is small, unstable, and easily swamped by a single aggressive print. It is not a swing-trading signal.

Aggressive buying/selling means marketable orders that lift offers or hit bids. Identifying aggressor side requires trade prints with side, or a reliable inference (Lee-Ready and cousins). OHLCV cannot measure aggressor flow. A green candle is not 'aggressive buying' in the microstructure sense; it is a net positive close.

4.2 Derivatives objects

Open interest is the stock of open contracts. Funding rate is the periodic payment rate. Predicted funding is a venue or model estimate of the next payment from current premium; treat it as an estimate that can move until the fixing. Liquidations are forced closures. Long/short positioning ratios, when published, are incomplete samples, not a census of all risk.

Basis is perp price minus spot (or mark minus index). Premium means perp rich to spot; discount means cheap. Basis is related to expected funding but is not identical to it; funding formulas use specified premium indices and clamps.

Crowded positioning is a hypothesis that one side has concentrated leveraged exposure. Observable proxies: extreme funding, rapidly rising OI with a one-way price move, visible liquidation clusters, crowded retail long/short ratios. None of these prove crowding of sophisticated books.

4.3 Useful combinations – and how they mislead

Combination	What it can describe	Misleading reading
Price + OI	Trend with adding exposure vs short-covering / long-liquidation rally	OI up + price up is not automatically healthy or sustainable
Price + Funding	Expensive vs cheap carry; possible crowding on one side	High positive funding is not a timed short signal
Price + OI + Funding	Whether a move is being paid for by the majority side	Three series can still be one story; not triple confirmation
Price + Volume	Participation in a move; possible exhaustion if new extrema print on falling volume	Fragmented crypto volume; falling volume is not a clock
Price + OI + Liquidations	Forced-flow contribution to a spike	Liquidation prints can be effects, not causes of the whole move

A common heuristic map: price up + OI up → new longs (and/or new shorts on the other side) adding; price up + OI down → shorts covering; price down + OI up → new shorts adding; price down + OI down → longs closing. This map is incomplete because both sides can add, transfers can occur across venues, and OI units / definitions differ.

HYPOTHESIS — NOT ESTABLISHED FACT

Extreme funding plus rising OI in the direction of the move increases the chance of a squeeze or flush later. Timing that later is a separate, unsolved problem. Many crowded markets stay crowded.

FAILURE / MISUSE

Never treat funding, OI, liquidations, or imbalance as deterministic. They are state descriptors. A cascade requires a path of forced orders through a finite book; that path is not observable in full.

WHEN USEFUL	WHEN UNRELIABLE	WHAT NOT TO INFER
Use derivatives state to estimate hold cost, crowding risk, and whether a move is likely to have forced-flow fuel.	Venue sample bias, cross-exchange leakage, predicted vs realized funding, and delayed liquidation tapes.	Do not infer that the other side 'must' be liquidated. Do not infer smart-money direction from funding sign.

SYSTEM RULE

If funding over the intended hold would consume a material fraction of planned edge (for example a large share of 1R), either shorten hold horizon, reduce size, or output NO TRADE. Carry is a cost, not a footnote.

5. Liquidity and Auction Behavior

5.1 What liquidity actually means

Liquidity is the capacity of the market to absorb size with limited impact and with reasonable certainty of fill. It has dimensions: tightness (spread), depth (size per level), resilience (how fast depth replenishes), and immediacy (ability to trade now). Volume is a cousin, not a synonym.

Types of liquidity

- Resting liquidity — limit orders in the book. Visible if the venue publishes them; still incomplete if iceberg/hidden orders exist.
- Aggressive liquidity — marketable flow willing to pay the spread to trade now.
- Stop liquidity — stop and stop-limit orders that become marketable after a trigger. Often invisible until triggered.
- Liquidation liquidity — forced orders generated by the venue risk engine when maintenance margin fails. Invisible as a book until they fire.

5.2 Grabs, sweeps, failed breakouts

A liquidity grab / sweep is a heuristic path: trade through a well-mapped level where stops may rest, then reverse. A failed breakout is the same family with a range-boundary definition. These are descriptions of a path that already happened. They become setups only if a reclaim rule and invalidation are coded.

5.3 Auctions, acceptance, rejection

Auction-market language says the market advertises a price and participants accept or reject it. Acceptance is time spent and volume printed in an area. Rejection is a quick excursion with little time/volume that returns. This is a model. It can be approximated with time-at-price and volume-at-price if those data exist; OHLCV approximates poorly.

5.4 Thin books, slippage, violent pockets

Crypto can move violently around liquidity pockets because: books are thinner than equity index futures; leverage is high; liquidations are automatic; markets are 24/7 with large variation in human and market-maker presence; venues are fragmented; and some alts have a few dominant LPs.

FACT

Impact is convex in size relative to depth. Doubling size more than doubles expected cost once you leave the near-touch. A system that backtests fills at mid or at the close will overstate edge.

5.5 What retail data can and cannot see

Question	Needs L2/L3 or venue internals	Can approximate from OHLCV / L1
Spread and touch size now	L1 minimum; L2 better	Not from closed candles
Impact for intended size	L2 depth snapshot (still stale)	Only a crude ATR/volume proxy
Aggressor imbalance	Prints with side or L3	Not from OHLCV
Stop clusters	Generally no (hidden)	Infer only via prior highs/lows — weak
Liquidation thresholds of others	No	Public liq tapes are after-the-fact
Sweep then reclaim	Price path; L2 optional	Yes, if sweep/reclaim is coded on bars

Question	Needs L2/L3 or venue internals	Can approximate from OHLCV / L1
Acceptance / time-at-price	Tick or minute bars help	Poor on coarse OHLC

SYSTEM RULE

If intended notional is large relative to visible near-touch depth or relative to recent volume, the setup is not executable as modeled.
Output NO TRADE or reduce size until impact is inside the cost budget.

6. Market Regimes

A regime is a temporary statistical climate: drift, volatility, correlation, and liquidity conditions that persist longer than a single setup but shorter than 'forever'. Strategies are usually regime-conditional. A rule that harvests trend drift will pay the range. A rule that fades extremes will pay the trend.

6.1 Regime catalog

Regime	Observable sketch	Typical strategy stress
Bull trend	HH/HL grammar, positive drift, dips bought	Short mean-reversion and fading breakouts
Bear trend	LH/LL grammar, negative drift, rallies sold	Long dip-buying without structure support
Sideways / range	Mean-reverting location, weak ADX, overlapping value	Breakout and trend-follow systems
High-vol	ATR/realized vol elevated vs own history	Stops too tight; size too large; fake levels
Low-vol	Compressed ATR, tight bands	Targets too far; breakouts stall; boredom trading
Trending volatility	Vol rising as trend extends	Mean-reversion and short-vol assumptions
Mean-reverting env.	Location extremes revert; weak drift	Momentum continuation
Breakout env.	Compression then directional expansion with volume	Range fades
Panic / liquidation	Vertical displacement, liq tape, wide marks vs last	Any system assuming continuity and stable correlations
Event-driven	Scheduled or shock news dominates	Technical setups that ignore event windows

6.2 Why strategies fail across regimes

A setup is a filter on paths. The distribution of paths changes when drift, vol, and liquidity change. Expectancy estimated in one climate is not portable. This is the main reason live results diverge from backtests even when execution assumptions are honest.

6.3 Detection using measurable variables

Prefer a small, coded classifier over a narrative label. Example feature set (illustrative, not a claimed edge):

- Volatility: ATR percentile, realized vol, band width, range/ATR.
- Trend strength: ADX, slope of a slow MA in ATR units, HH/HL vs LH/LL counts over a window.
- Volume: volume vs its N-period median; unusual prints.
- Derivatives: funding z-score, OI change rate, liquidation burst flags.
- Correlation: rolling BTC-alt return correlation; market breadth if a universe exists.
- Event flag: minutes to a scheduled catalyst.

HEURISTIC

A simple two-axis map — trend strength vs volatility — already prevents many mismatched trades. Complexity beyond that must earn its keep in out-of-sample tests.

SYSTEM RULE

Every setup type must declare compatible regimes. If the classifier is uncertain, or if the setup's compatible-regime flag is false, the default is NO TRADE, not 'try it anyway'.

WHEN USEFUL	WHEN UNRELIABLE	WHAT NOT TO INFER
Use regime as a gate on which setups are even eligible.	Regime labels flicker if features are too short-window. Too long-window and they lag regime breaks.	Do not infer the next regime from the current one. Persistence is a tendency, not a contract.

7. Setup Design

A setup is a fully specified mapping from market state to a candidate order, including the conditions under which the mapping outputs nothing. If any of preconditions, trigger, entry, stop, target, or invalidation is unspecified, it is not a setup. It is a vibe.

SYSTEM RULE

Setups are hypotheses about conditional expectancy. They are not assets. They are not 'the strategy'. No setup in this section is claimed to be profitable. They exist so the agent can evaluate completeness.

7.1 Anatomy of a setup

- Preconditions — regime, structure, liquidity, event window, data quality.
- Trigger — the event that makes the setup exist now rather than earlier.
- Entry — order type, price rule, size rule.
- Stop — invalidation price and order type.
- Target — one or more exits, including partials if used.
- Invalidation — conditions that cancel before entry or flatten after entry besides the stop.
- Expected regime — where the setup is even allowed to fire.
- Failure conditions — known ways the path usually dies.
- Required data — minimum fields. If missing, NO TRADE.
- NO-TRADE gates — explicit.

7.2 Worked templates (not recommendations)

A. Trend pullback

Preconditions: coded uptrend or downtrend (structure + optional ADX/MA-slope gate); volatility not in panic regime; no imminent high-impact event if the hold is short-horizon. Trigger: pullback to a predefined area (MA, prior HL, VWAP) with a momentum deceleration rule. Entry: break of pullback micro-structure or limit in the area. Stop: beyond the pullback extreme plus a buffer in ATR units. Target: prior swing or $k \cdot R$; trail only if that rule is pre-registered. Invalidation: trend grammar breaks before entry; depth too thin; funding consume-too-much rule. NO TRADE if: pullback has already traveled so far that R to structure invalidation is $<$ minimum; or the 'pullback' is indistinguishable from a regime break.

B. Breakout / retest

Preconditions: defined boundary; optional compression. Trigger: close beyond boundary with a volume or range expansion filter, then a retest that holds. Entry: on reclaim/hold of boundary. Stop: back inside the range beyond noise. Target: measured move or next mapped level. NO TRADE if: first break already extended many ATRs; boundary is a one-bar spike; event window; book is too thin to trust the break print.

C. Liquidity sweep / reclaim

Preconditions: an objective pool (equal highs/lows, PDH/PDL, session high) within a tolerance. Trigger: trade through the pool then close back through the level (reclaim) within N bars. Stop: beyond the sweep extreme. Target: opposite side of local range or mean. NO TRADE if: no reclaim, only a wick through; sweep occurs into a compatible trend-break rather than a range; stop-to-target R below minimum; high-vol panic regime where wicks are unstructured.

D. Range fade

Preconditions: range bounds stable for a minimum width and duration; low-to-moderate ADX; not immediately after a compression that may be a breakout regime. Trigger: tag of bound + rejection rule. Stop: beyond bound + buffer. Target: mid or opposite bound. NO TRADE if: bound has already been broken and accepted; upcoming event; R to mid is poor after spread.

E. Momentum continuation

Preconditions: expansion regime, directional displacement, participation. Trigger: shallow pause without structure break, then a new impulse. Stop: other side of the pause. NO TRADE if: first impulse is already extended; pause is actually a CHOCH; liquidity cannot support chase size.

F. Mean reversion

Preconditions: identified mean (session VWAP, range mid, slow MA) and a stretch statistic (z-score, %B, RSI) in a regime where stretch has historically reverted. Trigger: stretch + first sign of return. Stop: further stretch beyond a hard cap. NO TRADE if: trend-strength gate says the mean is moving away; stretch is news-driven; funding and OI suggest forced continuation.

7.3 Design rules

SYSTEM RULE		
If two setups fire in opposite directions on the same asset and timeframe, the output is not 'average them'. The output is NO TRADE unless a pre-registered priority rule exists (for example higher-timeframe regime vetoes the lower-timeframe fade).		
SYSTEM RULE		
Targets that exist only as hope are not targets. If the next objective level implies R below the system minimum after costs, the setup fails the gate.		
WHEN USEFUL	WHEN UNRELIABLE	WHAT NOT TO INFER
Use setups as checklists that force invalidation and cost to be explicit.	A template that is not parameterized and tested is not tradeable, even if it has a name.	Do not infer that a prettier chart is a higher-expectancy setup.

8. Risk Management

Risk management is the only part of a trading system that is under the operator's control on every trade. Edge is uncertain. Loss size can be bounded. If risk is not bounded, evaluation of edge is meaningless.

8.1 Risk per trade and position sizing

Risk per trade is the equity that will be lost if the planned stop is hit, including expected fees and slippage at exit. It is not margin posted and not notional.

```
risk_cash = |entry - stop| * size * contract_multiplier + expected_exit_costs
size = risk_budget_cash / (|entry - stop| * multiplier + per_unit_exit_cost)
```

Fixed fractional sizing risks a constant fraction f of current equity per trade. It automatically shrinks during drawdown and grows during equity expansion. f is a policy parameter. It is not derived from confidence theater.

HEURISTIC

Small f (often well under 1% of equity at stop) is how systems survive estimation error. Large f makes path variance dominate any plausible edge.

8.2 R-multiples, stops, targets, R:R

1R is the planned risk. A +2R winner made twice the planned risk. A -1R loser is a clean stop. A -1.4R loser means slippage, gap, or a stop that was not actually the risk model.

Stop placement belongs at invalidation, then size is derived. Reversing that — choosing size first and tightening the stop to make f look small — manufactures false precision and raises the rate of noise stop-outs.

Take-profit placement belongs at a pre-specified objective (structure, measured move, opposing liquidity) or a rule-based trail. Risk/reward before costs is $(\text{target} - \text{entry}) / (\text{entry} - \text{stop})$. After costs it shrinks. A '3R target' that never fills because it sits beyond typical excursion is decoration.

8.3 Expectancy and why win rate is not the goal

```
expectancy_R = p_win * avg_win_R + p_loss * avg_loss_R
expectancy_cash ≈ average(R) * average_risk_cash [only if risk is kept comparable]
```

A system with a 35% win rate and +2.5R average wins against -1R average losses has positive expectancy if those numbers persist after costs: $0.35 \cdot 2.5 + 0.65 \cdot (-1) = +0.225R$ per trade. A system with 80% wins of +0.3R and 20% losses of -2R has negative expectancy: $0.80 \cdot 0.3 + 0.20 \cdot (-2) = -0.16R$.

FACT

Win rate without payoff ratio is not a performance metric. Profit factor, expectancy, drawdown, and sample size are required before any quality claim.

8.4 Performance and risk statistics

Metric	Meaning	Use	Misuse
Expectancy	Mean outcome per trade in R or cash	Primary edge measure	Computing it on 12 trades
Profit factor	Gross profits / gross losses	Quick quality check	Ignoring sample and costs

Metric	Meaning	Use	Misuse
Max drawdown	Worst peak-to-trough equity path	Survival constraint	Treating past MDD as a cap that cannot be exceeded
Sharpe	Mean excess return / stdev of returns	Risk-adjusted comparison	Assuming normality; inflating with high-frequency overlapping returns
Sortino	Like Sharpe using downside deviation	Penalize harmful vol more	Still sample-sensitive
Calmar	Return / max drawdown	Path-quality snapshot	Unstable on short histories
Risk of ruin	Prob. equity hits a ruin barrier	Size policy	Using a toy formula that ignores fat tails and correlation
Consecutive losses	Clustered -R sequence	Circuit-breaker input	Believing N losses 'mean a win is due'

8.5 Portfolio, correlation, limits

Two longs on highly correlated alts are not two risks. They are one risk expressed twice. Aggregate by factor: BTC-beta, sector, direction, theme, and shared event exposure. Hard limits belong at: risk per trade, concurrent correlated risk, gross leverage, net directional exposure, daily loss, weekly loss, and strategy-level circuit breakers (disable a setup after a pre-specified degradation).

SYSTEM RULE

If adding a new position would push correlated exposure above the cap, the new position is NO TRADE even if it is locally beautiful. Portfolio risk dominates single-setup aesthetics.

SYSTEM RULE

Daily and weekly loss limits exist to interrupt process failure (bad data, bad regime classifier, runaway bot). They are not an opinion about the market's next move.

WHEN USEFUL	WHEN UNRELIABLE	WHAT NOT TO INFER
Use risk math to make trades comparable and survivable.	Metrics estimated on short, curated, or in-sample paths.	Do not infer skill from a high win rate or from a short Sharpe printed in a bull tape.

9. Statistics and Expectancy

The agent's job is not to admire a backtest number. It is to estimate whether a number is more consistent with a real conditional edge or with noise, bias, and search.

9.1 Core objects

Expected value of a trade distribution is the probability-weighted mean outcome. Expectancy in R is that mean expressed in planned-risk units. Variance and standard deviation describe dispersion. High expectancy with huge variance can still be untradeable at the chosen f .

Confidence intervals quantify uncertainty of an estimate given a model of sampling. They are only as honest as the independence and stationarity assumptions. Trading outcomes are rarely i.i.d. Treat textbook intervals as lower bounds on uncertainty, not as precision.

Sample size is the number of independent-enough trials. 30 correlated trades in one trend are not 30 independent tests of a setup. A rough system rule: do not promote a setup on a handful of trades; require a pre-declared minimum and still speak in probabilities.

9.2 Bias catalog the agent must name

- Selection bias — studying only setups that were taken or only markets that still exist in the dataset.
- Survivorship bias — dead coins, delisted contracts, failed venues drop out of the history.
- Data snooping / multiple testing — many rules were tried; the winner was kept. Naive p-values collapse.
- p-hacking — adjusting windows, filters, and exclusions until significance appears.
- Overfitting — the rule memorized noise. Symptom: excellent in-sample, collapse out-of-sample, brittle to small parameter changes.
- Underfitting — the rule is too crude to capture a real but conditional effect. Rarely the agent's main risk.
- Regression to the mean — extreme recent performance moves toward the less dramatic base rate.
- Base rates — most discretionary-looking patterns have no durable edge after costs. Start from skepticism.

9.3 Bayesian posture (process, not theater)

Let H be 'setup X has positive after-cost expectancy in regime R '. Prior $P(H)$ should be modest. Data D updates via $P(H|D) = \frac{P(D|H)}{P(D)}$. A pretty three-trade week is weak D . A pre-registered walk-forward that survives costs and parameter perturbation is stronger D . Conditional probabilities matter: $P(\text{win} | \text{sweep+reclaim, high-vol panic})$ is not $P(\text{win} | \text{sweep+reclaim})$.

HYPOTHESIS — NOT ESTABLISHED FACT

Many published trading patterns are artifacts of search over flexible charts. This handbook does not quantify that fraction because any specific percentage would be invented.

SYSTEM RULE

The agent may not claim an edge without: a pre-registered definition, cost-adjusted outcomes, an out-of-sample or walk-forward path, a sample-size caveat, and a statement of parameter sensitivity. Otherwise the correct language is 'insufficient evidence'.

9.4 Practical evaluation questions

- Is the mean R distinguishable from zero relative to its standard error, even under optimistic independence?
- Does the edge survive adding realistic fees, slippage, funding, and delay?
- Does it appear in more than one regime and more than one asset?
- Does a nearby parameter set still work, or is it a spike in parameter space?

- Was this rule found after looking at the same test set?
- What is the base rate of random entries with the same stop/target geometry?

10. Backtesting

A backtest is a simulation of a decision rule on historical data. It is not a time machine and it is not a live fill. Its job is to falsify bad rules cheaply.

10.1 Test types

Method	What it is	What it does not do
Historical backtest	Apply the rule to stored data	Guarantee future fills or stationarity
Train/test split	Fit on A, evaluate on later B	Fix look-ahead if features were built on all data
Holdout	A sealed final period	Help if the holdout was peeked at while iterating
Walk-forward	Roll train → test through time	Eliminate regime luck, but it reduces it
Out-of-sample	Data not used to choose the rule	Exist if the team 'just checked' it
Paper trading	Live data, simulated fills	Capture full live slippage or psychology of real size
Forward test	After freeze, observe	Replace statistical evaluation; it is one more sample

10.2 Costs and execution assumptions

- Transaction fees at the correct maker/taker tier — not an aspirational tier.
- Slippage model: at least spread + a size-vs-volume impact term. Mid-price fills are fantasy for takers.
- Funding paid or received over the actual simulated hold.
- Entry delay: signal on close of bar T, earliest action is T+1 unless using intra-bar logic that is actually implemented.
- Stop execution: assume adverse dead-through in fast markets; do not assume perfect stop fills at the stop price.
- Partial fills and rejected post-only orders if those order types are used.

FACT

OHLC cannot perfectly reproduce real execution. The high and low of a bar do not tell you the path, whether a limit would have been filled, or whether a stop would have slipped. Intra-bar assumptions are models. State them.

10.3 Common backtesting mistakes

- Lookahead: using the bar close to enter at the bar open; using future-normalized features; resampling leaks.
- Survivorship: testing only coins that are liquid today.
- Unrealistic fills on wick extremes.
- Ignoring funding and borrow.
- Optimizing on the same period used to report performance.
- Changing the rule after seeing the equity curve and keeping the same reported stats.
- Mixing timezones and candle alignments across venues.
- Assuming infinite liquidity on alts.

SYSTEM RULE

A setup may not be promoted from research to live eligibility until costs, delay, and stop-slip assumptions are explicit and a frozen evaluation period has been scored. Pretty in-sample curves are not eligibility.

11. Crypto-Specific Market Behavior

Crypto inherits microstructure from electronic futures and adds 24/7 trading, fragmented venues, retail leverage, uneven market-maker coverage, and reflexive narratives. Assumptions imported from equities or from BTC should not be pasted onto a mid-cap alt without modification.

11.1 Structural facts

- 24/7 markets — no official close. Session statistics are conventions. Weekend and holiday depth often thins.
- Fragmentation — price, volume, OI, and funding differ by venue. The agent's venue is the execution truth; others are context.
- Perpetual dominance — much speculative volume lives in perps, so funding and liquidation mechanics matter more than in cash-equity style analysis.
- BTC as risk factor — BTC dominance and BTC returns often drive alt beta. Dominance is the BTC share of reported crypto market cap; methodology varies by data vendor.
- Stablecoin rails — flows into USDT/USDC and depeg events can change collateral quality and risk appetite. A depeg is a collateral shock, not only a 'narrative'.

11.2 Correlation and beta

Alts often behave as high-beta expressions of BTC (and sometimes ETH) plus an idiosyncratic residual. In stress, residuals shrink and correlations rise. Three alt longs can become one BTC-beta long. ETH has periods of partial decoupling; that decoupling is itself unstable.

11.3 Asset-class assumptions

Asset class	Typical differences the agent must respect
BTC	Deepest books, tightest spreads, most robust index/oracle inputs, still cascade-prone at high leverage.
ETH	Usually second in depth; has idiosyncratic catalysts (upgrades, staking flows, L2 events).
SOL and large alts	Wider spreads, more venue-specific basis, more violent funding, thinner off-hours books.
Smaller alts	Impact dominates; OI can be concentrated; prints can move mark; data quality worse; gaps between venues larger.

11.4 Recurring crypto climates

Funding regimes cluster: crowded long carry in risk-on tapes; short squeeze funding flips; alt-specific extreme funding when OI is large vs inventory. Weekend behavior is a heuristic of thinner books and occasional gap-like expansions, not a law. Liquidation cascades are more central than in unlevered cash markets. Catalysts include halvings, ETF flows, exchange incidents, protocol exploits, unlocks, listings, and regulatory actions. Each must be treated as an event flag, not as an automatic direction.

SYSTEM RULE

Default size and maximum leverage must be asset-class dependent. A BTC-sized notion of 'normal ATR' applied to a thin alt is a data error disguised as a setup.

12. Macro

Crypto trades inside a global risk-asset complex. Macro variables can shift the discount rate, dollar liquidity, and risk appetite that feed BTC-beta. That does not make the agent a macro fund.

12.1 Variables that can matter

Variable	Why it can matter	How not to use it
Policy rate / path (Fed and others)	Changes cash yield vs speculative duration and leverage cost	Do not predict FOMC from candles
Treasury yields	Discount-rate and liquidity proxy	Do not assume a fixed daily beta to 10Y
DXY	Dollar strength often inverse to risk and to crypto in some regimes	The correlation changes; it is not a trading signal by itself
CPI / PCE / inflation prints	Shift real-rate expectations	The print ≠ the market response; surprise matters more than the headline
Employment / NFP	Labor data moves policy odds	Same surprise logic
FOMC / minutes / QT-QE language	Direct liquidity and guidance shock	Event-window risk; direction is not known from the calendar alone
VIX / risk-on-off	Cross-asset risk appetite	Crypto can diverge; VIX is not a crypto indicator
Geopolitics	Risk shocks, energy, sanctions, operational disruption	Do not invent a geo-to-BTC mapping after the move

12.2 Incorporation without becoming a forecasting machine

SYSTEM RULE

Macro enters the agent as context flags and risk gates, not as a predicted next candle. A scheduled high-impact event can justify NO TRADE or reduced size. A macro narrative must not override a missing micro setup or a broken invalidation.

HEURISTIC

Surprise versus consensus usually matters more than the raw number. If the agent does not have a consensus series, it should not pretend to score surprise.

Useful operational pattern: maintain an economic calendar; mark high-impact windows; freeze or de-risk discretionary setups through the window unless the setup is specifically an event-volatility rule that was tested as such.

13. News and Event Analysis

News is information that may change other participants' willingness to hold risk. Most news does not produce a durable directional edge for a retail-frequency agent. The main skill is not storytelling. It is refusing false causality.

13.1 Event types

- Scheduled: CPI, FOMC, options expiry, known unlocks, announced governance votes, listed conference talks.
- Unexpected: exploits, outages, regulatory emergency actions, sudden ETF flow shocks, wars, exchange failures.
- Market-structure events: listings/delistings, fee changes, leverage-cap changes, insurance-fund events.
- Asset-specific: token unlocks, partnerships, protocol upgrades, validator incidents.

13.2 NEWS EXISTED vs NEWS CAUSED THE MOVE

FACT

Temporal coincidence is not causation. Price can move before a headline (leak, positioning, unrelated flow), with a headline (liquidity air-pocket plus attention), or after a headline (slow digestion). A model that always attaches the nearest headline to the nearest candle is a narrative generator, not an analyst.

Minimum causal hygiene:

- Timestamp the news independently of the candle the agent wishes to explain.
- State what was already in the price: prior trend, funding, OI, and calendar.
- Ask whether similar headlines historically moved this asset, in which direction, and for how long — if such a sample exists.
- Allow the residual: 'unexplained flow' is a valid conclusion.
- Never write 'because' from a single joint occurrence.

SYSTEM RULE

The agent may record 'headline present' as context. It may not assert 'headline caused the move' unless a pre-specified causal protocol is satisfied. Default language: 'news present; causal contribution unknown'.

FAILURE / MISUSE

LLMs are especially prone to narrative fallacy here. If price is already known, the model will find a story. Hide the future path when asking the LLM to assess a headline's plausible impact.

WHEN USEFUL	WHEN UNRELIABLE	WHAT NOT TO INFER
Use news as a risk window and as a feature flag for regime (event-driven).	Unverified social posts, recycled headlines, and timestamps without timezone.	Do not infer that a bullish headline requires a long, or that a selloff 'must' have been caused by the item on the feed.

14. Multi-Timeframe Analysis

A timeframe is a sampling rate. The 5m, 15m, 1h, 4h, and 1D charts are the same trade tape at different resolutions. They are not independent markets and they are not independent confirmations.

14.1 Roles

Timeframe	Typical role in a systematic stack
1D	Slow regime and major structure. Poor for precise execution.
4h	Swing structure, broader invalidation, hold-horizon context.
1h	Common decision timeframe for intra-day to multi-day perps setups.
15m	Trigger refinement and nearer invalidation.
5m	Execution timing only if the setup's edge was tested there. Easy to overfit.

A coherent design: higher timeframe sets regime and maximum invalidation; lower timeframe sets trigger and tighter stop only if that tighter stop still represents true invalidation rather than noise.

14.2 Conflicts

Conflicts are normal: 1D up, 15m down is a pullback or a break. The agent does not 'resolve' this by stacking indicators until they agree. Agreement across nested sampling rates is partly tautological.

SYSTEM RULE

Do not treat three timeframes agreeing as three pieces of evidence. Prefer one primary decision timeframe, one higher-timeframe gate, and optionally one lower-timeframe execution rule. If gates conflict and no priority rule exists, NO TRADE.

FAILURE / MISUSE

Scanning every timeframe for a story is how discretionary traders manufacture activity. An AI will do the same if prompted to 'find confluence'.

15. Correlation and Portfolio Risk

Risk aggregates. The book does not care that tickets have different tickers if they share the same driver.

15.1 Mechanisms

- BTC/ETH correlation: often high, unstable in the tails.
- Alt beta: residual idiosyncratic vol exists in calm periods and can vanish in stress.
- Hidden exposure: long SOL, long an SOL-ecosystem token, and long an alt-beta basket is one crowded risk-on expression.
- Long/short concentration: net delta, plus the risk that both legs gap the wrong way if correlation snaps.
- Stress correlation: pairwise estimates from calm windows understate cascade co-movement.

FACT

Three trades can be one trade. If their returns share a dominant factor, adding tickets does not add diversification. It adds the illusion of diversification and a larger hole if the factor moves.

SYSTEM RULE

Before entry, compute incremental portfolio delta in BTC-equivalent and in correlated-cluster units. If the new ticket does not change the book in a controlled way, it is either redundant (NO TRADE) or it is an unintentional stack.

15.2 Practical aggregation

Maintain clusters (BTC, ETH, large-cap beta, mid-cap beta, idiosyncratic). Cap risk per cluster and gross risk across clusters. Re-estimate correlations on a rolling window and apply a stress overlay (assume correlations move toward 1 in liquidation regimes).

16. Trading Psychology and Behavioral Finance

This section is not a pep talk. Humans leave fingerprints on order flow. System designers also leak human bias into parameters, prompts, and override habits. The agent must understand both.

16.1 Biases that show up in markets

Bias	Market fingerprint (heuristic)	System defense
FOMO	Chase after vertical displacement; late breakouts	Extension cap; no entry after $k \cdot \text{ATR}$ from trigger
Revenge trading	Immediate reverse after a stop	Cooldown; flatten-and-halt rules
Loss aversion	Widening stops; refusing to exit	Stops pre-committed; no LLM authority to move them
Confirmation bias	Seeking agreeing indicators / headlines	Pre-registered features only
Recency bias	Overweight last week of PnL	Frozen evaluation windows; base-rate tables
Overconfidence	Size up after wins; skip gates	Fixed f ; hard caps
Gambler's fallacy	Expecting a win after N losses	Each trial independent unless a coded dependence exists
Narrative bias	Forcing a story onto a tape	Causal hygiene in Part 13
Anchoring	Stuck on a round number or entry	Mark-to-market vs invalidation, not vs entry pride
Herd behavior	Crowded funding and one-way OI	Crowding as risk, not as permission
Fear / greed	Regime of urgency or euphoria	Vol and funding flags; reduce size, do not moralize

16.2 Implications for an AI agent

The model can reproduce every human bias in text form: urgency, revenge, narrative completion, overconfidence after a fluent paragraph. Fluency is not evidence.

SYSTEM RULE

The LLM may not increase size, widen stops, add 'just this once' overrides, or invent a setup because the tape 'feels' like it is getting away. Those actions require a coded policy change under version control, not a paragraph.

SYSTEM RULE

Human override channels, if they exist, must be logged as such. An override is not a setup outcome and must not poison expectancy statistics without a flag.

17. Machine Learning and AI in Trading

17.1 What LLMs are good at

- Reading unstructured text (news, documentation, governance posts) and extracting entities, times, and claims.
- Comparing a structured setup packet against this handbook's checklists.
- Writing an auditable thesis from features the code already computed.
- Summarizing journals and clustering qualitative tags after numeric analysis exists.
- Flagging missing fields ('invalidation not provided').

17.2 What LLMs are bad at

- Arithmetic they are not given tools for: indicators, size, liq price, expectancy.
- Faithful recall of exact prices, timestamps, and tables without the data in context.
- Resisting narrative completion when a price path is visible.
- Being a stationary policy: prompts drift; temperature and sampling add instability.
- Knowing what they do not know. Confidence language is cheap.

Failure modes specific to LLM traders

Hallucination — invented levels, invented papers, invented 'usual win rate of this pattern'. Pattern interpretation without a definition — seeing a pattern that is not coded. Feature leakage / lookahead — using language that implies knowledge of the next bar. Prompt drift — silent instruction changes that alter policy. Data contamination — context that already contains the evaluation period. Overfitting in prose — a verbal rule that is just as curve-fit as a numeric one.

17.3 Required division of labor

Layer	Owns
CODE	Data ingestion, clocks, candles, indicators, features, setup detection, sizing, entry/stop/target arithmetic, monitoring, fills, journals, statistics, evaluation, circuit breakers.
LLM	Interpret structured context, compare candidate setups, qualitative news/docs, produce a thesis, explain a decision, summarize evidence, recommend NO TRADE when the packet is incomplete.
HUMAN / GOVERNANCE	Risk limits, strategy versions, go-live permission, capital, venue credentials.

SYSTEM RULE

The LLM is not allowed to calculate technical indicators or invent price levels. If a level is not in the structured packet, it does not exist for decision purposes.

SYSTEM RULE

If structured evidence and LLM narrative disagree, structured evidence wins or the output is NO TRADE. A fluent disagreement is not a vote.

18. The No-Trade Framework

NO TRADE is the default. A trade is an exception that survived gates. The agent should emit NO TRADE with reasons, the same way it would emit an order ticket.

18.1 Hard gates (any one is sufficient)

- Data invalid: missing candles, stale quotes, desynced clock, duplicate bars, implausible prints.
- Setup incomplete: no coded trigger, no invalidation, no size from risk budget.
- Risk/reward after costs below minimum R.
- Stop inside noise (stop distance below a $k \cdot \text{ATR}$ floor or inside a multiple of spread).
- Intended size exceeds depth/impact budget.
- Leverage / liquidation distance too close to stop or to a mapped cascade zone.
- Event window flagged high-impact and the setup is not an event strategy.
- Regime mismatch for the setup type.
- Conflicting setups with no priority rule.
- Correlated exposure cap would be breached.
- Daily or weekly loss limiter tripped, or strategy circuit breaker open.
- Funding / carry would consume a material fraction of modeled edge over intended hold.
- Historical evidence missing, stale, or negative for this setup times regime cell.
- Model disagreement: LLM thesis requires a level or fact not in the packet.
- Uncertainty high: classifier confidence or data quality below threshold.
- Unclear invalidation: if you cannot state the stop, you cannot trade.

18.2 Soft gates (stack toward NO TRADE)

Several mild defects can sum to a refusal: mediocre R, mediocre depth, mildly elevated vol, noisy structure, weak volume, moderate event risk. Do not barely pass four soft gates and call it confluence.

SYSTEM RULE

When uncertain, output NO TRADE. When two actions are close, prefer smaller size only if size reduction is a pre-registered policy. Do not invent a half-setup.

OUTPUT SCHEMA

Every decision should include: decision in {TRADE, NO_TRADE}; reasons[]; gates_failed[]; setup_id; packet_hash; timestamp; data_quality_ok; regime_label. A TRADE without this envelope is a process failure.

19. Trade Journaling

A journal is the dataset on which the agent learns. If a field is not stored, it cannot be used later to distinguish bad luck from bad process. Unstructured prose is a supplement, not a substitute.

19.1 Required fields for every idea

Group	Fields
Identity	idea_id, timestamp_utc, venue, asset, direction, timeframe, setup_id, strategy_version
Decision	TRADE / NO_TRADE, reasons, confidence (calibrated score, not poetry)
Geometry	entry, stop, targets[], planned_R, size, notional, leverage, margin_mode, est_liq
Costs	expected_fees, expected_slippage, expected_funding_over_hold
Features	technical feature vector, structure labels, indicator values actually used
State	regime, ATR, realized_vol, volume stats, spread, depth snapshot summary
Derivatives	funding, predicted_funding, OI, OI_change, recent_liquidations summary, basis if available
Context	news flags, macro flags, event window, correlation / portfolio snapshot
Thesis	structured thesis text, invalidation text, what would upgrade/downgrade confidence
Execution	order types, fill prices, realized slippage, partials, rejects, latency
Outcome	exit, realized_R, fees, funding_paid, PnL, MFE, MAE, time_to_target, time_to_stop
Quality	data_quality flags, whether stop was true invalidation vs noise, override flag

MFE (maximum favorable excursion) is the best unrealized R while the trade was open. MAE (maximum adverse excursion) is the worst. Together they tell you whether targets and stops are geometrically sane. Losers that never went +0.3R are different from losers that were +1.8R first.

SYSTEM RULE

NO TRADE ideas are journaled too. Otherwise the dataset is conditioned on the agent already having liked the tape.

19.2 Why structure matters

Learning requires grouping: setup times regime cells, cost-adjusted R, stability over time. Prose cannot be averaged. If the LLM writes the thesis, store the packet that produced it so explanations remain auditable.

20. Learning from Losses

A loss is not a verdict on the entire strategy. It is one sample. Classification of losses prevents the two failure modes of 'change everything after one stop' and 'learn nothing forever'.

20.1 Loss taxonomy

Class	Meaning	Typical action
Bad thesis	Preconditions were false or ignored	Process bug; do not tweak parameters on one case
Correct thesis, bad timing	Direction eventually right, path killed the stop first	Study MAE; maybe execution timing – only with a sample
Correct direction, poor entry	Filled at an extended price	Entry rule review
Stop too tight	MAE exceeded stop but MFE reached target path	Numeric counterfactual first
Unexpected news	Event not in calendar / unforecastable shock	Event-gate review; not all shocks are preventable
Regime mismatch	Setup fired in a climate it was not built for	Classifier or gate bug
Liquidity event	Gap, cascade, book pull	Size and asset-class policy
Random variance	Inside the historical loss distribution	No rule change
Execution problem	Slippage, reject, delay, wrong order type	Ops fix
Data problem	Stale/missing/wrong candle or mark	Halt and repair

20.2 Counterfactuals — numeric first

The LLM may explain a counterfactual only after code computes it. Examples of computable questions:

- If entry were delayed 15 minutes / 1 bar, what would R have been?
- If stop were 1 ATR wider, would planned target have been reached before the wider stop? What would the historical false-positive rate of that wider stop be?
- If the same setup in a different regime cell is selected, what is the historical expectancy of that cell?
- What were MFE and MAE in R at fixed checkpoints?

SYSTEM RULE

Do not change a stop rule because a wider stop would have saved the last trade. Compute the effect of that wider stop on the whole tagged sample, including the winners it would have worsened.

FAILURE / MISUSE

Hindsight language is the default LLM mode after a loss. Require a class label from the taxonomy plus numbers before any prose post-mortem.

21. Learning and Rule Evolution

The agent may discover conditional regularities from its journal. It may not rewrite itself after every scar. Learning is a promotion process, not a reflex.

21.1 Learned-rule examples (illustrative form)

- Setup X: mean R is worse when funding z-score exceeds a threshold.
- Setup Y: mean R is better when volume exceeds its 20-period median.
- Setup Z: expectancy is negative in low-vol range regime.

These sentences are research tickets. They become gates only after a protocol: definition freeze, minimum sample, out-of-time check, and version bump.

21.2 Safeguards

- Minimum sample per cell before a learned rule may even be proposed (pre-declare; do not invent a universal N here as science).
- Multiple-testing control: many candidate conditions will look good by chance.
- No peeking at the frozen evaluation period while searching.
- Parameter changes create a new strategy_version. Old version remains queryable.
- Frozen evaluation periods: after promotion, live or paper results accrue without mid-period edits.
- Decay: learned rules expire if not revalidated.

SYSTEM RULE

The AI must not rewrite strategy logic after a single loss or a short losing cluster. Clusters can trip a circuit breaker (halt) without constituting a rewrite.

21.3 Versioning

A version is a hash of: setup definitions, gates, sizing policy, feature code, and prompt template IDs. Journals reference the version. Comparisons across versions are how drift is detected.

22. Edge Detection

An edge is a repeatable, cost-adjusted, out-of-sample tendency for a defined decision rule to produce positive expectancy with acceptable path risk. It is not a feeling, a narrative, or a good week.

22.1 Vocabulary the agent must not confuse

Term	Meaning
A good trade	A trade that followed the process: valid packet, gates passed, planned risk honored.
A profitable trade	A trade with R greater than 0. Many bad-process trades are profitable. Many good-process trades are not.
A good strategy	A versioned rule set with evidence of positive expectancy and survivable drawdown across evaluation protocols.
A lucky streak	A short path whose mean exceeds what the estimated distribution would usually deliver.
A genuine edge	Positive expectancy that survives costs, sample scrutiny, regimes, and parameter perturbation, with no identified fatal bias.

22.2 Evidence checklist

- Positive expectancy in R after fees, slippage, funding.
- Consistency: not one month and zeros elsewhere, unless that month is a declared regime the rule targets.
- Robustness: nearby parameters and nearby assets do not invert the sign.
- Out-of-sample / walk-forward performance.
- Multiple regimes or an honest admission of regime-specificity plus a detector.
- Stability of the equity path: drawdown compatible with capital policy.
- Sample size large enough that the mean is not a toy.
- Low enough parameter sensitivity that the rule is describable.

SYSTEM RULE

If any box is empty, the agent must say 'edge unproven' rather than 'edge present'. Unproven is the normal state. Trading while unproven is a capital decision, not a language decision, and must be labeled research-size if it happens at all.

23. Failure-Mode Catalogue

Each row is a way the agent can look systematic while transferring money to the tape or to costs.

Mode	What it is	Why it happens	Damage	Detect	Prevent
Overfitting	Rule matched noise in sample	Searching many variants	Live collapse	In/out gap; brittle params	Freeze defs; walk-forward; penalize complexity
Lookahead bias	Future info in past decision	Close-of-bar misuse; feature leak	Fantastical backtests	Bar-timing audit	Signal at T, act at T+1 unless intra-bar is real
Survivorship bias	Only winners remain in data	Delistings dropped	Overstated alt edges	Compare vs point-in-time universe	Keep dead markets in research data
Data leakage	Test labels seep into training	Shared preprocessing	Fake skill	Pipeline review	Strict time cuts
Selection bias	Studying only taken trades	Journaling winners	Blind to missed costs	Log NO TRADE too	Idea-level dataset
LLM hallucination	Invented facts/levels	Fluent completion	Trades from fiction	Grounding checks	Packet-only levels; refuse extras
Narrative fallacy	Story after the tape	Causal hunger	False lessons	Timestamp discipline	Part 13 protocol
Excessive trading	Too many tickets	No activity penalty	Death by costs	Turnover vs edge	NO TRADE default; cost hurdle
Strategy drift	Live rule != tested rule	Silent tweaks	Uninterpretable PnL	Version hash mismatch	Change control
Prompt drift	Policy text changes	Unversioned prompts	Unstable decisions	Prompt ID in journal	Treat prompts as code
Regime change	Climate left the sample	Nonstationarity	Edge disappearance	Detector disagreement; vol shift	Gates; halt on unknown regime
Transaction costs	Fees ignored or understated	Optimistic tier	Negative real EV	Live vs modeled fee gap	Use actual fee schedule
Slippage	Fills worse than model	Mid-price fantasy	Size illusion	Realized vs expected slip	Impact model; size caps
Funding costs	Carry ignored	Short-horizon bias in tests	Slow bleed	Hold-time attribution	Charge funding in EV
Correlation	Stacked same bet	Ticker diversity theater	Amplified drawdown	Factor exposure report	Cluster caps
Small sample	Noise dressed as research	Impatience	Random promotion	Error bars	Pre-declared N; stay humble
Testnet distortion	Testnet is not mainnet	Different liquidity/behavior	False readiness	Divergent fills	Treat testnet as integration only
Bad / stale / missing / duplicate data	Wrong world state	API gaps, retries, clocks	Wrong signals, double events	Candle integrity checks	Halt on quality fail
API failures	Cannot see or cannot cancel	Venue or network	Unmanaged risk	Heartbeat / reject spikes	Flatten or halt policy
False confidence	Verbal certainty	LLM style; recent PnL	Oversizing	Calibration vs outcomes	Confidence is a scored field

SYSTEM RULE

A detected failure mode is a halt or a versioned fix, not a prompt that says 'please avoid this next time'.

24. Hyperliquid-Specific Knowledge

VERIFICATION REQUIRED

Hyperliquid contract specs, fee tiers, leverage caps, API fields, and exact mark/oracle formulas change. The statements below are based on Hyperliquid public documentation as commonly published (oracle, mark, funding, liquidations, contract specifications pages). Re-read the live docs before wiring production logic. Do not invent undocumented API behavior.

24.1 Product shape

Hyperliquid perpetual markets are linear-style perpetual contracts without expiry. Funding tethers them to an underlying reference. Documentation describes USDC-margined contracts with oracle prices often denominated in USDT for assets whose primary spot liquidity is off-platform, and USDC denomination for certain assets whose primary spot liquidity is on Hyperliquid (docs have cited examples such as HYPE-USD and PURR-USD). That quanto-like detail matters for interpretation of USD PnL. Verify current specs.

Accounts can use cross or isolated margin. Initial margin fraction depends on user-selected leverage. Documentation states maintenance margin fraction as half of the maximum initial margin fraction, with tiered maintenance on some assets. Verify the live margin tables.

24.2 Oracle price

FACT

Per Hyperliquid docs: validators publish spot oracle prices for each perp asset on a short cadence (docs: approximately every 3 seconds). Oracle is used in funding and is a component of mark. Validators compute a weighted median of specified CEX and (conditionally) Hyperliquid spot mids. Weights and included venues are documented and can change. Assets with primary spot liquidity on Hyperliquid may exclude external sources until liquidity thresholds are met; assets with primary liquidity elsewhere may exclude Hyperliquid spot. The clearinghouse oracle is a stake-weighted median of validator submissions.

24.3 Mark price

Docs describe mark as a robust perp fair value used for margining, liquidations, TP/SL triggers, and unrealized PnL. A documented construction is the median of: (1) oracle plus a 150-second EMA of (Hyperliquid mid minus oracle); (2) median of best bid, best ask, last trade on Hyperliquid; (3) a weighted median of specified CEX perp mids. Additional fallbacks apply when some inputs are missing. Mark updates with oracle publications. Exact current formula: verify docs.

FAILURE / MISUSE

Mark can differ from last trade and from mid during violence or thin books. Stops that trigger on mark can fire when last trade has not printed the same price. Backtests on last-price OHLC will not reproduce mark-triggered events.

24.4 Funding

Documented mechanics (verify before coding):

- An interest-rate component is specified for consistency with CEXs: 0.01% per 8 hours, i.e. 0.00125% per hour, economically a carry paid toward shorts when premium is neutral.
- Premium component from perp vs oracle using impact prices, not a single last trade.
- Documented 8-hour-style formula: $F = \text{average premium } P + \text{clamp}(\text{interest} - P, -0.0005, +0.0005)$.
- Premium sampled frequently (docs: every 5 seconds) and averaged over the hour.
- Funding is paid every hour at one-eighth of the 8-hour-style rate.

- Payment uses $\text{position_size} * \text{oracle_price} * \text{funding_rate}$ — docs emphasize oracle, not mark.
- A high cap exists (docs have stated 4% per hour). This is an extreme bound, not a normal rate.
- HIP-3 / builder-deployed perps may use a different premium formula. Do not assume one formula for all markets.

24.5 Liquidations

Liquidations use mark. Docs note that during high volatility, mark may differ substantially from book price. Estimated liquidation prices shown at order-entry can differ from post-trade estimates because book liquidity and cross-margin uPnL change. A documented isolated-form sketch is:

$$\text{liq_price} = \text{price} - \text{side} * \text{margin_available} / \text{position_size} / (1 - l * \text{side})$$

with $\text{side} = +1$ long / -1 short, and $l = 1 / \text{maintenance_leverage}$, with extra complexity on tiered assets and cross margin ($\text{margin_available} = \text{account_value} - \text{maintenance_margin_required}$). Verify before implementing a replica calculator. Treat venue-reported liq price as authoritative when available.

24.6 Book, execution, fees, slippage

Hyperliquid uses an on-chain order book. Market, limit, stop, TP, and other order types exist in the product surface; exact time-in-force and trigger semantics must be read from current API/docs. Fees are tiered by rolling volume and can include staking discounts and maker rebates. Do not hard-code a fee tier from memory. Pull the live schedule.

Slippage is not a published constant. It is a function of size vs book at send time plus competing flow. A documented market-order helper in some integration docs uses a slippage bound; treat any such bound as an implementation detail to verify.

24.7 Testnet vs mainnet vs data-only testing

Mode	What it can validate	What it cannot
Historical market-data replay	Signal logic, indicators, journals	True mark/funding path if not stored; true fills
Testnet trading	API signing, order state machine, error handling	Mainnet depth, fee economics, cascade behavior, honest slippage
Mainnet paper or tiny size	Closer data identity	Still not full impact at target size
Mainnet execution	Reality	Still not a large enough sample to prove edge

SYSTEM RULE

Do not assume testnet fills, OI, funding, or liquidation behavior match mainnet. Promotion criteria must include mainnet-data evaluation of costs and a staged size ramp.

24.8 Data the agent should subscribe to (conceptual)

- Trades, L2 book, mid, mark, oracle, funding, predicted funding if provided, OI, 24h stats.
- User fills, order acks/rejects, position, margin, liq price, funding payments.
- Integrity: sequence gaps, stale heartbeats, clock offset vs venue time.

Exact endpoint names and payloads: official API docs only. This handbook will not invent them.

25. System Decision Framework

Run these questions in order. Stop at the first hard failure and emit NO TRADE. Do not skip ahead to thesis prose because the chart is interesting.

#	Question	Fail action
1	Is the data valid (complete, fresh, internally consistent)?	NO TRADE; halt if persistent
2	What regime label does the classifier emit, and is it confident?	NO TRADE if unknown / low confidence
3	What coded setups are present on the decision timeframe?	NO TRADE if none
4	Are those setups historically supported in this regime cell?	NO TRADE if evidence missing or negative
5	What is current volatility vs the setup's operating band?	NO TRADE if outside band
6	Is liquidity sufficient for planned size?	Reduce size only if policy allows; else NO TRADE
7	What do funding and OI imply for hold cost and crowding?	NO TRADE if carry or crowding gate fails
8	Are there major news or macro windows?	NO TRADE or mandatory de-risk if flagged
9	Are signals conflicting across setup types or timeframes?	NO TRADE unless a priority rule exists
10	What is the proposed entry (order type and price rule)?	NO TRADE if unexecutable as specified
11	Where is invalidation, and is it economically a real invalidation?	NO TRADE if unclear
12	What is expected R after costs?	NO TRADE if below minimum
13	Does it pass hard risk filters (f, cluster cap, daily loss, liq distance)?	NO TRADE
14	What does the historical cell say (mean R, N, MDD, sensitivity)?	NO TRADE if unproven at this size
15	Does the LLM thesis use only packet facts?	If it invents, discard thesis or NO TRADE
16	Is calibrated confidence above threshold?	NO TRADE
17	Is there a reason to wait (incomplete bar, upcoming funding, pending event)?	WAIT / NO TRADE
18	Is there any remaining reason to refuse?	Prefer NO TRADE over a low-quality ticket

SYSTEM RULE

Preference order: NO TRADE greater than wait greater than small research-size if explicitly in research mode greater than full-size trade. Low-quality trades are not better than nothing. They destroy the journal and the equity path.

26. Reference Tables

Quick retrieval. Details live in the numbered parts. Do not use these tables as standalone trading rules.

26.1 Indicators

Concept	Meaning	Useful for	Common misinterpretation
SMA / EMA	Smoothed price location	Location vs mean; lagging drift	Independent confirmation of each other
VWAP	Volume-weighted session mean	Intraday location, execution bench	Universal support
RSI	Scaled up vs down mass	Momentum coordinate	Overbought = short now
MACD	Fast-slow EMA gap	Drift / momentum family	Zero cross as prophecy
ATR	Typical range scale	Stops, size, regime	Direction
Bollinger	Mean +/- k sigma	Location and compression	Squeeze gives direction
Stochastic	Close in N-bar range	Range position	Always fade 80/20
ADX	Directional strength	Regime gate	High ADX = long
OBV / volume	Activity and signed activity	Participation context	Sees aggressor side

26.2 Market structure

Concept	Meaning	Useful for	Common misinterpretation
Swing H/L	Parameterized pivot	Structure map	Unique objective truth
HH HL LH LL	Swing grammar	Trend language	Guarantees continuation
BOS / CHOCH	Coded structure breaks	Invalidation	Holy grail reversal clock
S/R, zones	Prior interest areas	Candidate levels	Must hold
Sweep / grab	Through-then-reclaim path	Mean-revert family setups	Stops were stolen as a plan
PDH/PDL, sessions	Calendar extrema	Mapped liquidity	Magic daily levels

26.3 Derivatives metrics

Concept	Meaning	Useful for	Common misinterpretation
Funding	Periodic long/short payment	Carry, crowding context	Always fade extreme funding
Predicted funding	Estimate until fix	Near-term carry	Locked-in rate
OI	Open contract stock	Add vs flatten	OI up is bullish
Basis / premium	Perp vs spot/index	Tether stress	Fair-value oracle of next trend
Liquidations	Forced closes	Cascade risk after the fact	A scheduled fuel gauge you can time
Mark / oracle	Venue reference prices	Liq, uPnL, funding base	Same as last trade

26.4 Risk metrics

Concept	Meaning	Useful for	Common misinterpretation
Risk per trade	Cash lost at stop + costs	Sizing	Margin posted
R-multiple	Outcome / planned risk	Comparability	Quality of thesis by itself
Expectancy	Mean outcome	Edge estimate	A promise
Profit factor	Gross win / gross loss	Quick screen	Works on tiny N
Max DD	Worst path	Survival	A ceiling that cannot break
Sharpe / Sortino / Calmar	Risk-adjusted snapshots	Comparison	Proof of skill on short samples
Risk of ruin	Barrier probability	f selection	Exact in fat tails

26.5 Statistical concepts

Concept	Meaning	Useful for	Common misinterpretation
Sample size	Independent-enough trials	Humility about means	30 correlated trades = 30 tests
Confidence interval	Estimate uncertainty (model-based)	Not over-reading a mean	True 95% in non-i.i.d. data
Multiple testing	Many tries inflate winners	Research protocol	The best variant is the truth
Overfitting	Noise memorized	Complexity control	More filters = more edge
Base rate	Unconditional frequency	Bayesian prior	Ignored because the chart is vivid
Regression to mean	Extremes soften	After hot streaks	Next trade is due to lose

26.6 Setup types

Concept	Meaning	Useful for	Common misinterpretation
Trend pullback	Enter with trend after pause	Trend regimes	Any dip is a pullback
Breakout/retest	Acceptance beyond a bound	Expansion regimes	First wick through is enough
Sweep/reclaim	Stop-run then return	Range / mapped liquidity	Every wick is a sweep setup
Range fade	Reject a bound	Mean-reverting regimes	Fade a breaking market
Momentum continuation	Add after shallow pause	Expansion	Chase a completed impulse
Mean reversion	Stretch back to a mean	Stationary location	Fade a regime shift

26.7 Failure modes (short)

Concept	Meaning	Useful for	Common misinterpretation
Lookahead	Future in the past	Backtest audit	More data always helps
Hallucination	Ungrounded claims	LLM control	Fluent = correct
Cost blindness	EV before fees/funding	Promotion gates	Small costs do not matter
Correlation stack	One factor, many tickets	Portfolio caps	More tickers = diversified
Stale data	Acting on yesterday's book	Quality halt	Last candle is always fresh

26.8 Regime types

Concept	Meaning	Useful for	Common misinterpretation
Bull / bear trend	Persistent directional grammar	Allow trend setups	Must continue tomorrow
Sideways	No persistent grammar	Allow fades	Safe, low risk
High / low vol	Dispersion climate	Size and stop policy	High vol = good trading
Panic / liq	Forced-flow climate	Halt or hard de-risk	Great opportunity by default
Event-driven	Calendar or shock dominates	Gate technical setups	Direction is known from the calendar

26.9 News categories

Concept	Meaning	Useful for	Common misinterpretation
Scheduled macro	Known time, unknown surprise	Risk window	Headline direction = trade direction
Regulatory	Rule or enforcement shock	Regime flag	Always instantly priced correctly
Exploit / outage	Operational shock	Halt asset or venue	Buy the dip algorithmically
Unlock / listing	Supply or access change	Asset-specific flag	Always sell unlocks
ETF / flow news	Wrapper demand	BTC-beta context	Linear mapping from headline to ticks

26.10 Macro indicators

Concept	Meaning	Useful for	Common misinterpretation
Policy rate / FOMC	Cost of money, guidance	Event gates	Forecast the statement from RSI
CPI / PCE	Inflation prints	Surprise windows	High CPI always dumps crypto
NFP / labor	Labor surprise	Same	Always risk-off
DXY / yields	Dollar and discount-rate context	Risk climate	Fixed hedge ratio
VIX	Equity-implied fear	Cross-asset mood	A crypto signal

27. AI Retrieval Format and Operating Contract

27.1 How to retrieve

Query by concept name plus one of: definition | calculation | information content | interpretation | failure | interaction | required data | NO-TRADE | FACT | HEURISTIC | HYPOTHESIS | SYSTEM RULE. If a retrieved section conflicts with live venue docs, venue docs win for mechanics.

27.2 Packet the LLM should receive

- Identity: asset, venue, ts, timeframe, strategy_version.
- Data-quality block.
- Regime block with feature values, not only a label.
- Candidate setups with coded fields populated.
- Book/cost block: spread, depth summary, fee tier, funding, OI.
- Portfolio block: cluster exposures, daily P&L vs limits.
- Event block: calendar + news flags without a forced causal story.
- Historical cell: N, mean R, MDD, last validation date — or none.

27.3 Language constraints

SYSTEM RULE

Forbidden outputs: invented statistics, invented levels, 'smart money', guaranteed targets, causal 'because' from a single coincidence, size increases from excitement, stop widening from hope.

SYSTEM RULE

Required outputs on every call: TRADE or NO_TRADE; reasons; which handbook rules fired; what would change the decision. If evidence is weak, say so in those words.

27.4 Additional knowledge an agent still needs (outside this book)

- Live Hyperliquid docs and API schemas.
- The system's own coded setup specifications and parameter files.
- The system's own journal statistics.
- Current fee tier, balances, positions, open orders.
- Clock synchronization and venue maintenance calendar.

28. Execution, Data Integrity, and Operations

A thesis that cannot be sent, acknowledged, amended, or canceled is not a trade. Operational risk is a first-class risk factor.

28.1 Clocks, candles, and alignment

FACT

A 1h candle is defined by a timezone and an epoch alignment. UTC vs exchange-local vs daylight-saving shifts will silently desynchronize session VWAP, PDH/PDL, and event windows. The agent must freeze a timezone policy (usually UTC) and store it on every bar and every ticket.

28.2 Order-state machine

Know the difference between: submitted, accepted, rejected, open, partially filled, filled, canceled, cancel-rejected, trigger-set, trigger-fired. A stop that is not acknowledged is not a stop. A flatten that is rejected is an emergency.

SYSTEM RULE

No new risk may be added unless flatten/cancel paths are confirmed healthy (heartbeat, rate limits, successful recent cancel or a synthetic health check). If the agent cannot exit, it must not enter.

28.3 Integrity checks

- Monotonic timestamps; no duplicate trade IDs; no overlapping candles.
- Mark/oracle/last sanity bands (implausible jumps vs prior mark without a print).
- Sequence IDs or equivalent for websocket streams; gap then resnapshot.
- Reconciliation: local position vs venue position on a short interval.
- Unit checks: base vs quote qty, sign of shorts, decimal scales.

28.4 Rate limits, latency, queue

Bursting the API can delay the only cancel that matters. Prioritize reduces and cancels over new entries. Latency is part of slippage. A signal that assumes zero delay is a lookahead cousin.

28.5 Kill switches

- Loss limits (daily/weekly).
- Data-quality halt.
- Max position, max leverage, max orders outstanding.
- Unexplained inventory (position without a ticket).
- Venue-status halt (maintenance, degraded matching).
- Manual human halt always wins.

28.6 Stablecoin and collateral risk

USDC-margined perps inherit stablecoin depeg and transfer-rail risk. A collateral event can move markets and can change the value of margin simultaneously. Treat depeg flags as event-driven regime, not as a clever relative-value toy, unless that is a separately specified strategy.

28.7 What this handbook still cannot replace

Legal constraints, custody design, key management, and capital allocation are governance. The agent should assume it is not allowed to raise its own limits.

SYSTEM RULE

If any operational check fails, the decision is NO TRADE at best and HALT at worst. Do not work around a broken feed by interpolating candles.

29. Closing Constraints

The market does not owe the agent a trade. Most moments are not setups. Most setups are not edges. Most edges, if they exist, decay. The only durable advantages available by policy are: not inventing trades, not lying to the journal, not ignoring costs, not stacking correlated risk, and not rewriting rules on a scar.

SYSTEM RULE

Final preference: NO TRADE over low-quality trade. Unproven edge over claimed edge. Coded invalidation over narrative invalidation. Venue documentation over this handbook whenever mechanics disagree.